



INFINEON TECHNOLOGIES AG
Internal Correspondence

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Subject: PSoC Edge 84 Security Software Limitations
Category:
Distribution:

1. SUMMARY

The following memo describes the architectural and customer impact on having a secure processing environment (SPE) running on the same core (CM33), simultaneously with a non-secure processing environment (NSPE). It will explain the impact of the NSPE when the SPE is executing, what changes would need to be implemented by Infineon to make it a full proof solution as well as the next steps and proposals to the customers and internal teams.

2. THE PROBLEM

As mentioned in the summary, the key requirement to having a secure system on the PSoC Edge 84 is to be able to drive real-time behavior as well as ensuring secure services are available when required. The Expedition platform was designed in a way to have a Secure Partition Manager.

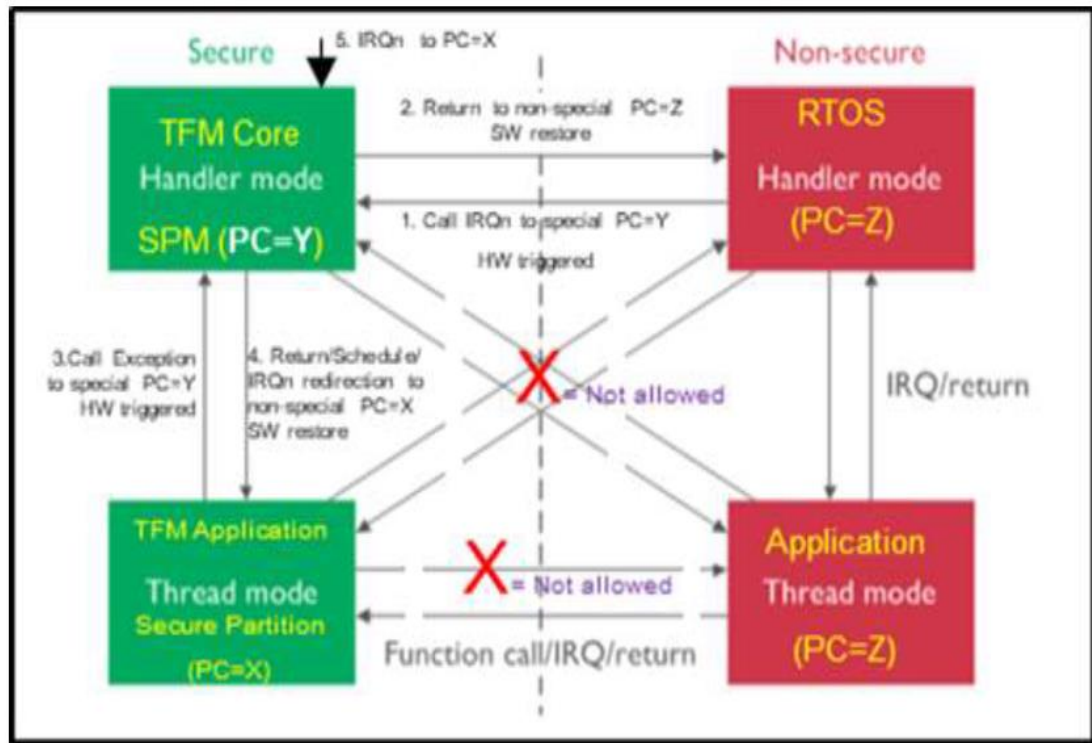


Figure 5 – Context Switch & Pre-emption Model w/TZ and PCs

The following context switches are available within TF-M for PSoC Edge 84

Table 13 - Supported Context Switches

	TF-M Context Switches – Manage PCs Across Them	Supported
1	SPE => NSPE: SPE is pre-empted by a NSPE interrupt	No
2	NSPE=>SPE: An NSPE exception handler returns to the pre-empted SPE	No
3	NSPE=>SPE: NSPE is pre-empted by an SPE interrupt	No
4	SPE=>NSPE: An SPE exception handler return to pre-empted NSPE	No
5	NSPE=>SPE: NSPE makes a Client API function call into SPE (IPC model)	Yes
6	SPE=>NSPE: SPE returns a Client API call from NSPE (IPC model)	Yes
7	SPM=>SP: SPE-SPM schedules SPE-Secure Partition during initialization	Yes
8	SP=>SPM: SPE-Secure Partition makes Secure Partition API calls into SPE-SPM	Yes
9	SPM=>SP: Return from SPE-SPM call into SPE-Secure Partition (return from SPM call)	Yes
10	SPE=>NSPE: SPE makes a function call into NSPE	No
11	NSPE=>SPE: NSPE returns a call from SPE	NO



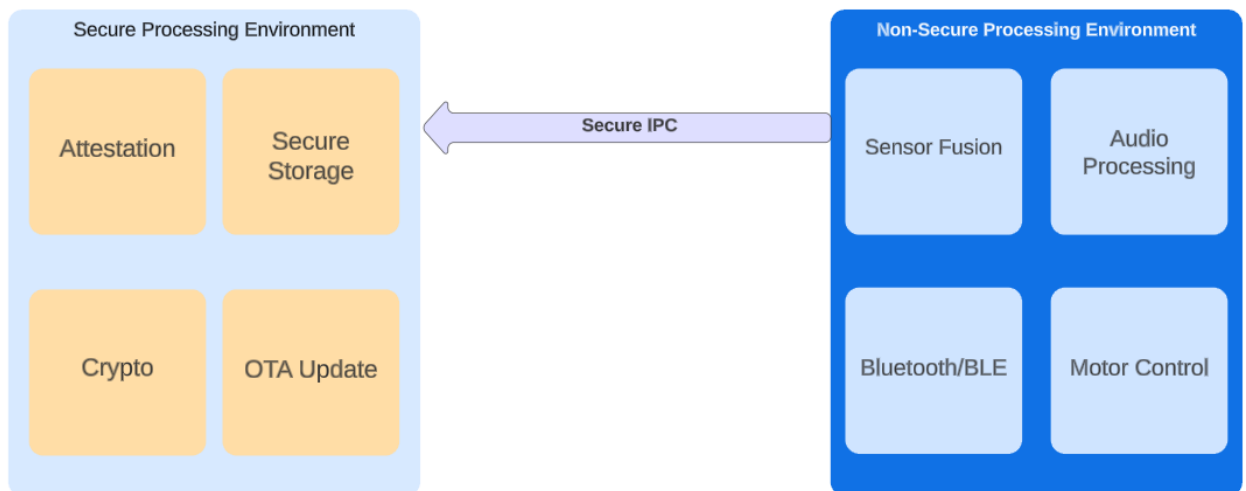
As set out in the Security Architecture Specification, it determines that for simplicity, the NSPE cannot interrupt the SPE within the TF-M context. This decision has a massive knock-on effect of the control the NSPE requires to execute its real-time requirements to run its business logic and timing related sensor capture.

- To simplify TF-M and Protection Context usage, SPE pre-emption by NSPE interrupts shall be disabled with a TF-M configuration option as illustrated in Figure-2.
- NSPE interrupt pre-emption shall not block watchdog timer event to prevent system hang or DoS.

To understand the consequences of this, here are some examples of the NSPE needing timing related priority to ensure the functions of the product:

- Bluetooth timing requirements of the stack
- Collecting sensor data from the DMA and interrupts
- Processing voice input ensuring samples aren't dropped.
- Performing motor control functions

A simple functional diagram for a secure system is shown below.



What this ultimately means is when context is switched to the SPE, context cannot be passed back to the NSPE until the SPE runs to completion. The stock TF-M implementation does not have any isolated timing requirements, however, when NSPE requests a service from the SPE within the CM33, the NSPE CM33 will be blocked until completion resulting in a synchronous blocking call to the SPE.



It is not known the timing requirements of each service for L2 and L4 part (requesting Secure Enclave or CM33 SPE services would result in different timing).

Based on the current trajectory and requirements, customers will not be able to run a SPE and have full timing control of interrupts in the NSPE to make a truly low power, secure and real-time system. This means customers will have to make the following choices:

1. Ensure a TF-M service is only called when not requiring time critical interrupts or tasks within the NSPE. Depending on the time it takes to execute a service, they may have to stop functionality.
2. Dedicate the CM33 core to be an SPE if there needs to be real-time execution and secure services simultaneously. This will have an effect on the low-power execution of the device.
3. Putting all the real-time code into the SPE (Trusted Firmware-M). This will require product makers to have engineers experienced in TF-M, potentially violate best practices in the security world (leaner is more secure) and make code less portable or increase maintenance cost.

In all cases except bullet two, the CM55 would have to avoid accessing the SPE directly and pass through the CM33 NSPE to avoid the M55 starving the NSPE of context on the CM33.

In the case of a wearable, key functions like Bluetooth, sensor fusion, Wi-Fi connectivity could all use services like Attestation, OTA, crypto or secure storage that will ultimately block the core functions until executed in the SPE.

3. LONG TERM SOLUTIONS

The following section describes the high-level options to solve the problem of forcing the customer to make difficult decisions about their architecture outlines above. These solutions only solve the interrupt priorities but not the context switching for higher priority tasks.

1. Have all interrupts handled by the Secure Processing Environment. This is a similar solution to option three in section 2 of the memo and forces the developer to be exposed developing in the SPE.
2. Set the NSPE interrupts to secure, which may have some security implications.

For task switching, it's not clear what the solution is, however, the RTOS needs to be aware of the SPM to be able to hand context back to the secure world (or keep it for a higher priority NSPE tasks). This is more complex and needs a deeper dive to understand.



4. NEXT STEPS AND PROPOSAL

Before moving towards a solution, the following needs to be achieved:

1. Have a customer committed to using TF-M who is not willing to take the approaches detailed in section two.
2. To do a competitive analysis with STM and NXP to understand if they have the same limitations when running a SPE and NSPE on the same core.
3. Create or add to an existing application note the trade-offs and details of using a secure processing environment and the run-time consequences. This should make recommendations depending on the use cases of the customers.
4. Obtain the absolute timing when executing the provided secure services within TF-M. This will allow for better guidance to customers on how long the NSPE is going to be blocked for. This can help guide the customer on what DMA buffer sizes (to cover the time the interrupt isn't services).
5. Based on point three and four, create or add to an existing application note the learnings to help customers make good design decisions
6. Get some realistic proposals on how to solve situations in the table for NSPE/SPE interrupts and task priorities including protection context switching.
7. Ensure Marketing content is updated to relay the design choices and considerations to ensure customers are aware of any limitations or run-time behavior

Education is key when interacting with customers and it's imperative that when we are driving security solutions with our customers, that they know the impact and consequences of any architectural decision. Without this information, customers could be lead down a path that ultimately delays, or even worse, prevents them getting to market.